

***Suggested Teaching Guidelines for
Natural Language Processing & Computer Vision
PGCP-AI August 2026***

Duration: 120 Hours: 40 Theory (T) + 60 Lab (L) + 20 Self-Learning (SL)

Objective: To Introduce the students to Natural Language Processing & Computer Vision.

Prerequisites: Good knowledge of Python Programming and fundamentals of Artificial Intelligence.

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment – 20%

List of Course material

Text Book:

Natural Language Processing, by Pushpak Bhattacharyya, Aditya Joshi / Wiley India, 2023

Reference Book:

1. The Big Book of NLP Techniques by Shlomo Vaknin
2. Theory and Practice of NLP Coaching: A Psychological Approach by Bruce Grimley

Note:

- **Lab assignments are indicative; faculty shall provide adequate assignments for better practice.**

I.NLP (26 Hours Theory + 42 Hours Lab + 15 Hours Self Learning)

Session 1: (2T + 2L + 1SL)

Theory and Lab

Language in Cognitive Science

- Definitions of language
- Language and symbolic systems: Artificial language (Logical language/programming Language) vs. Natural Language
- Linguistics as a scientific study

Session 2: (2T + 2L + 1SL)

Theory and Lab

Language Analysis and Computational Linguistics

- Language Analysis:
 - Paradigmatic and Syntagmatic relationship
 - Form, Function and Meaning in Language Analysis
 - Levels of Linguistic Analysis: Phonetics, Phonology, Morphology, Syntax, Semantics,
 - Discourse, Pragmatics, Lexicology
 - Artificial Intelligence (AI) and its sub disciplines
 - Natural Language Understanding (NLU)
 - Natural Language Generation (NLG)
 - Natural Language Interaction (NLI)

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Assignment: Difference between Semantics and Pragmatics

Session 3: (2T+ 4L+ 1 SL)

Theory

Shallow Parsing and Tools for NLP

- Morphological Analysis
- Tokenization & PoS Tagging
- Chunking & Multi word expression (MWE)
- Named-Entity Recognition
- Lemmatizer & Stemming
- Morphological Synthesis
- Word Sense Disambiguation
- Universal Networking Language

Lab Assignment: Apply tok , stemming , lemma, and POS tagging for following data
<https://www.kaggle.com/arushchillar/disneyland-reviews>

Session 4: (2T+ 4L + 1SL)

Theory and Lab

Deep Parsing and Tools for NLP

- Syntactic Parsing Techniques and algorithms
- Semantic Parsing
- Information Extraction
- Automatic Summarization
- Anaphora Resolution, Pragmatics and Discourse analysis
- Ontology & Semantic Web

Session 5: (2T+ 4L + 1SL)

Theory and Lab

Statistical Approaches

- Probability Theory & Models
- Discrete Time Models
- Markov Models, Entropy Models
- Statistical Parsing
- Text Categorization / Classification and Clustering
- Text Classification Using Support Vector Machine (SVM)
- Centroid based Classification

Session 6: (2T + 2SL)

Theory and Lab

NLP with Machine Learning and Deep Learning

- Machine Learning & Deep Learning for NLP
- ML vs. DL along with respective algorithms names
- Linear Algebra Review
- Long Short-Term Memory (LSTM)

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Session 7: (2T+ 4L+ 2SL)**Theory and Lab****Study of Important ML/DL Algorithms & Case Studies:**

- Pre-processing
- Need of Pre-processing Data
- Introduction to NLTK, spaCy
- Using Python Scripts

Session 8: (2T + 5L + 1SL)**Theory**

- Introduction to gates, GRU and LSTM
- Introduction to LLM
- Hallucinations and Limitations of LLMs

Assignment –Lab:

- Implement LSTM and GRU in python
- Load a pretrained Large Language Model using Hugging Face to perform text generation, summarization, and question answering

Session 9: (2T + 5L + 1SL)**Theory and Lab**

- Sequence Modeling - Word representation
- word embedding matrix
- learning words and embedding's
- Sentiment classifications
- Introduction to Prompt Engineering
- Learn how to design effective prompts for summarization

Session 10: (2T + 4L + 2SL)**Theory and Lab****Word2Vec models (Skip-gram, CBOW, Glove, one hot Encoding)**

- Sequence-to-sequence models (Seq2Seq) - GloVe: Global Vectors for Word representation
- Brief on Sequence-to-Sequence Models
- RNN based Sequence-to-Sequence Model
- Challenges
- Transformer in NLP
- Understanding the Model Architecture
 - Getting Hang of Self- Attention
 - Calculation of Self-Attention
 - Limitations of the Transformer
- Transformer-XL o Using Transformer for Language Modeling
- Using Transformer- XL for Language Modeling

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Session 11: (2T + 4L + 1SL)

Theory and Lab

- Bert in NLP
- Model Architecture
- BERT Pre-Training Tasks
- NLP Model Deployment Techniques using Flask

Sessions 12 & 13: (4T + 4L + 1SL)

Theory

Speech Processing

- Articulatory Phonetics
- Speech Sounds and Phonetic Transcription
- Acoustic Phonetics
- Phonology
- Computational Phonology
- Digital Signal Processing Techniques
- Automatic Speech Recognition (ASR)
- Speech Recognition Approaches
- Text to Speech (TTS) system
- Speech Synthesis Approaches
- Language Models

Lab

NLP Applications

- Lexicon, Dictionaries, thesaurus
- Transliteration, Spell Checker, Grammar Checker, Domain identification, Language identification
- Auto suggest/ Auto complete, Gender Prediction
- Machine Translation
- Neural machine translation
- Information extraction and Retrieval
- Question answering & dialogue agents
- Speech Technologies
- OCR, Hand Writing Recognition
- Indian Language Script Technology
- Language identification
- Auto suggest/ Auto complete, chat bots, Robotics

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II. Computer Vision (14 Hours Theory + 18 Hours Lab + 5 Hours Self-Learning)

Sessions 14 & 15: (4T + 4L + 1SL)

Theory

- Introduction to Computer Vision
- Computer Vision and Natural Language Processing
- The Three R's of Computer Vision
- Basics of Image Processing
- Vision Level – Low, Mid & High

Lab Assignments: Load any .jpeg image and process the image using Open CV

Sessions 16 & 17: (4T + 4L + 1SL)

- Edge Detection, Interest Points and Corners
- Image Classification
- Recognition, Bag of Features, and Large-scale Instance Recognition
- Transfer Learning

Lab Assignments: Load any image and classify the image and implement the edge detection

Session 18: (2T + 6L + 1SL)

- Alex Net,
- ResNet,
- Image Net
- Introduction to Vision Transformers (ViT)
- Image Patch Embeddings, Self-Attention in Computer Vision

Lab Assignments: Implement AlexNet, ResNet and Image Net on the data sets.
Implement Image Classification using a Pretrained Vision Transformer (ViT)

Sessions 19 & 20: (4T + 4L + 2SL)

Introduction to Object Detection Algorithms

- YoLo (High Level Overview)
- RCNN
- Fast RCNN
- Faster RCNN
- Mask RCNN

Lab Assignments:

Applications of Computer Vision

- Gender Prediction
- Image Mapping
- Face / Object Recognition
- Image Recreation
- Image Matching